

Ishpreet Nagi

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EDUCATION

McMaster University

Bachelor of Applied Science, Computer Science

Hamilton, ON

Sept 2021 — May 2026

- Cumulative GPA: Summa Cum Laude | Dean's Honor List
- Relevant Coursework: Data Structures, Algorithms, Databases, Operating Systems, Distributed Systems, Concurrency, Computer Architecture, Cryptography, Discrete Mathematics

WORK EXPERIENCE

Software Engineer and Research Coordinator

June 2024 — July 2025

Healthcare Systems Research & Analysis

Waterloo, ON

- Designed and built distributed backend data pipelines in **Python** with **PostgreSQL**, **Pandas**, and **NumPy** to ingest, preprocess, and validate large interconnected datasets, **automating 90% of downstream deployment**
- Implemented RESTful service interfaces and background job workers for long-running data processing, with retry logic, schema validation, and clear technical documentation for downstream consumers
- Tuned relational queries and schema design (indexing, joins, batch reads) to keep multi-million-row workloads responsive, applying system-design principles from first principles rather than ORMs alone
- Directed a team of 4 engineers across 8 stakeholders, driving architecture decisions, code reviews, and on-time delivery of every milestone

Machine Learning Analyst and Research Assistant

May 2023 — May 2025

McMaster University - Department of Computing and Software

Hamilton, ON

- Built and maintained backend evaluation and model-serving infrastructure in **Python** under Dr. Douglas Down, achieving **10% accuracy gains** while shrinking model footprint by 15%
- Independently maintained a research codebase across 2+ years — versioned schemas, reproducible pipelines, CI checks, and clear docs — enabling other researchers to extend it without hand-holding

PROJECTS

Kollec App (kollec.app)

Sept 2025 — Apr 2026

- Designed and shipped scalable REST APIs and microservices using **Next.js API routes**, **Prisma ORM**, and **PostgreSQL**, owning the full backend lifecycle from schema design through production deployment
- Built a stateful peer-to-peer trade-matching engine backed by complex relational queries and asynchronous data updates across users, collections, and offers, applying DS&A choices to keep response times low at scale
- Implemented query optimization and indexing strategies (composite indexes, pagination, N+1 elimination) and added structured logging so slow paths are visible in production
- Instrumented an event-driven workflow for asynchronous card-identification jobs, hardening the service with retries, idempotency keys, and technical docs for internal API consumers
- Built and maintained CI/CD pipelines with **Git**, **GitHub Actions**, and **Docker** within a team of 7, driving code review and test-coverage standards

Technical Executive, DeltaHacks 10 (github.com/deltahacks/landing-10)

Aug 2023 — May 2024

- Independently maintained the DeltaHacks 10 platform over a full cycle, shipping to **1000+ applicants** under a hard deadline
- Built instrumentation and structured activity logging for user engagement, treating telemetry as a first-class backend concern

Personal Website (ishpreetnagi.com)

July 2024 — Present

- Built and self-hosted a responsive portfolio using Astro and TypeScript, achieving a **95+ Lighthouse score** across all metrics

SKILLS

- **Languages:** Python, Java, TypeScript, JavaScript, SQL (PostgreSQL, MySQL), C/C++ (familiar)

- **Backend & APIs:** REST, RPC (familiar), microservices, Node.js, Next.js API routes, Spring Boot (familiar), Prisma ORM, event-driven architecture
- **Data Stores:** PostgreSQL, MySQL, Supabase, Redis (familiar), Elasticsearch (familiar), query optimization, schema design, indexing
- **Systems & Infra:** Distributed systems, concurrency, system design, technical documentation, Kafka/Flink (interest), AWS (interest), Docker, GitHub Actions (CI/CD), Git
- **AI in the Dev Loop:** GitHub Copilot, Cursor, Claude Code, OpenAI API, Claude API, LLM integration